

C. Maximum Even Sum

time limit per test: 2 seconds
 memory limit per test: 256 megabytes

You are given two integers a and b . You are to perform the following procedure:

First, you choose an integer k such that b is divisible by k . Then, you simultaneously multiply a by k and divide b by k .

Find the greatest possible **even** value of $a + b$. If it is impossible to make $a + b$ even, output -1 instead.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains two integers a and b ($1 \leq a, b \leq a \cdot b \leq 10^{18}$).

Output

For each test case, output the maximum **even** value of $a + b$ on a new line.

Example

input	Copy
7	
8 1	
1 8	
7 7	
2 6	
9 16	
1 6	
4 6	
output	Copy
-1	
6	
50	
8	
74	
-1	
14	

Note

In the first test case, it can be shown it is impossible for $a + b$ to be even.

In the second test case, the optimal k is 2. The sum is $2 + 4 = 6$.

Codeforces Round 1047 (Div. 3)

Finished

Practice



→ Virtual participation

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Start virtual contest

→ Submit?

Language: Python 3.13.2
 Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
337420409	Sep/07/2025 19:20	Accepted

→ Problem tags

brute force greedy implementation math
 No tag edit access

→ Contest materials

- Announcement (en) 